

Perhaps one of the most riveting statements on the cloud by Marc Benioff, the founder of Salesforce, aptly describes the potential of cloud in the world of IT. With the explosion of cloud and serverless compute, there has also been an explosion of creative software, as developers are no required to buy expensive and high maintenance infrastructure to run their applications but instead can simply loan computing power on demand from one of the providers.

‘Cloud’ refers to the general internet whereas ‘Cloud services’ refer to all the services that are provided over the internet on-demand. While earlier this has been limited mostly to things like email servers, a new brand of services targeted specifically at software development and deployment has been gaining traction.

‘Cloud computing’ is defined by Wikipedia as an information technology (IT) paradigm, a model for enabling ubiquitous access to shared pools of configurable resources (such as computer networks, servers, storage, applications and services), which can be rapidly provisioned with minimal management effort.

This means that instead of worrying about infrastructure and non-functional aspects of an application, a developer can now concentrate on the core logic leaving the resources to be managed by the cloud service which can be accessed on-demand.

Google Cloud has a rather large number of services, ranging from a simple VM machine running Linux to a video analysis and intelligence API that can identify the context of most videos within minutes.

To use Google Cloud, we need to sign up for an account that among other details, necessitates payment information (usually via credit card). Once done, we are given some free credits to try our hand at the services offered or play around with them.

Let us take a look at the two most popular Google Cloud services.

Compute Engine

Google Compute Engine allows you to create scalable Virtual Machines (VMs) of different capacities in a matter of minutes. It provides a clear and easy API to setup a VM using only a few commands, for example:

```
• gcloud compute instances create my-vm --custom-cpu 4
  --custom-memory 5
```

Would create a VM with 4 CPUs and 5GBs of RAM memory.

Let us try to create a VM on Google Cloud and connect to it.